

Jonathan Colaço Carr

jonathan.colacocarr@mail.mcgill.ca | j-c-carr.github.io | linkedin.com/in/jonathan-colaco-carr

Education

McGill University/Mila , M.Sc in Computer Science	2023 – 2026
• Supervisors: Prakash Panangaden and Doina Precup	
Stanford University , Visiting Graduate Student	2024 – 2025
• Supervisor: Benjamin Van Roy	
McGill University , B.Sc in Honours Mathematics	2018 – 2023
• Minors in Computer Science and Physics	

Awards

FRQNT Doctoral Scholarship , \$100,000	Apr 2026
Four-year scholarship for Canadian students, determined by academic record and research potential.	
MITACS Globalink Research Award , \$6,000	Sep 2024
Research award for Canadian graduate students studying abroad, determined by expected research quality.	
McGill Graduate Mobility Award , \$8,900	Sep 2024
Scholarship for McGill graduate students studying abroad, determined by academic standing.	
NSERC Canada Graduate Scholarship – Master’s , \$17,500	May 2023
Graduate scholarship determined by academic excellence, research potential, and interpersonal skills.	
McGill Computer Science Undergraduate Research Award , \$7,000	May 2022
Research award determined by academic record and research aptitude.	
McGill Space Institute Summer Undergraduate Research Award , \$7,000	May 2021
Research award determined by academic record and extracurricular leadership.	
RBC x Microsoft AI for Social Impact Challenge, 2nd Place Prize , \$2,000	Apr 2020
National AI competition prize determined by originality, technical excellence, impact, and feasibility.	

Publications

* denotes co-first authorship.

- [1] **J. Colaço Carr**, “An analysis of optimization targets for value alignment within the Markov occupancy problem,” M.S. thesis, McGill University (in press), 2026, [pdf].
- [2] **J. Colaço Carr**, P. Panangaden, D. Precup, and B. Van Roy, “Reinforcement learning with pairwise preferences in long-term decision problems,” in *International Conference on Machine Learning (ICML)*, 2026, acceptance rate: 26.6%, [pdf].
- [3] **J. Colaço Carr**, Q. Sun, and C. Allen, “Focused skill discovery: Learning to control specific state variables while minimizing side effects,” in *Reinforcement Learning Journal (RLJ)*, 2025, acceptance rate: 39.0%. [pdf], [code].
– Preliminary version presented as an extended abstract at RLDM 2025.
- [4] K. Chehbouni*, **J. Colaço Carr***, Y. More, J. C. Cheung, and G. Farnadi, “Beyond the safety bundle: Auditing the Helpful and Harmless dataset,” in *Proceedings of the Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL)*, 2025, acceptance rate: 22.6%, **oral presentation**. [pdf], [code].
– Preliminary version selected for a spotlight presentation at the NeurIPS 2024 workshop on Algorithmic Fairness through the Lens of Metrics and Evaluations.

- [5] **J. Colaço Carr**, P. Panangaden, and D. Precup, "Conditions on preference relations that guarantee the existence of optimal policies," in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024, acceptance rate: 27.6%. [pdf].
- [6] J. Kennedy, **J. Colaço Carr**, S. Gagnon-Hartman, A. Liu, J. Mirocha, and Y. Cui, "Machine-learning recovery of foreground wedge-removed 21-cm light cones for high- z galaxy mapping," in *Monthly Notices of the Royal Astronomical Society (MNRAS)*, 2024, [pdf].

Research Experience

- Mila (Reasoning and Learning Lab)** May 2022 – present
- Supervisors: Prakash Panangaden and Doina Precup
 - Used human models from economics to design sequential decision-making algorithms aligned with human goals.
 - Audited a human preference dataset for label quality, topic coverage, and changes in LLM behaviour after fine-tuning.
- Stanford University** Oct 2024 – Mar 2025
- Supervisor: Benjamin Van Roy
 - Extended the maximal lottery social choice rule to sequential decision-making problems and designed game-theory algorithms for computing solutions. Implemented experiments in JAX.
- UC Berkeley (Center for Human-Compatible AI)** Jun 2024 – Sep 2024
- Supervisors: Cameron Allen and Stuart Russell
 - Developed a simple method to improve skill discovery using state abstraction, benefiting both efficiency and safety.
- McGill Space Institute** May 2021 – Dec 2021
- Supervisor: Adrian Liu
 - Implemented computer vision algorithms to recover signals from noisy telescope data.

Work Experience

- Hortus AI**, Lead Software Developer Dec 2024 – Dec 2025
- Led a team of developers (15 in total) to build a consumer reports platform for AI tools. Reported directly to startup founder, Thomas Krendl Gilbert. Platform now used by over 100 public officials across the United States.
 - Designed algorithms to rank and evaluate AI tools in collaboration with the GovAI coalition.
- The Democratic Engagement Exchange**, Pilot Researcher Aug 2025 – Oct 2025
- Conducted a feasibility study on the use of personalized AI assistants for voter education in Canada.
- McGill University**, Teaching Assistant Jan 2024 – Apr 2024
- Held weekly office hours and created assignments for a graduate-level course in machine learning (COMP 551).
- AltaML**, Machine Learning Intern Jan 2022 – Apr 2022
- Implemented machine learning algorithms to identify cross-selling opportunities.

Extracurricular Experience

- McGill AI Safety x Law Group**, Co-founder Apr 2022 – Dec 2023
- Co-founded and taught an eight-week course on AI safety for McGill Law students.
- RBC x Microsoft AI for Social Impact Challenge**, Team Lead Sep 2019 – Apr 2020
- Led a four-student team to design a textile recycling robot for an AI social impact challenge. Placed 2nd out of 167 teams from across Canada.
- The McGill Tribune**, Editorial Board Member Apr 2019 – Apr 2020
- Collaborated with journalists and graphic designers to publish articles for McGill's student newspaper.
- NASA Space Robotics Challenge (Phase 2)**, Team Lead Jul 2019 – May 2020
- Led a five-student team to design simulated localization and mapping algorithms for a space robotics competition.